

HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026



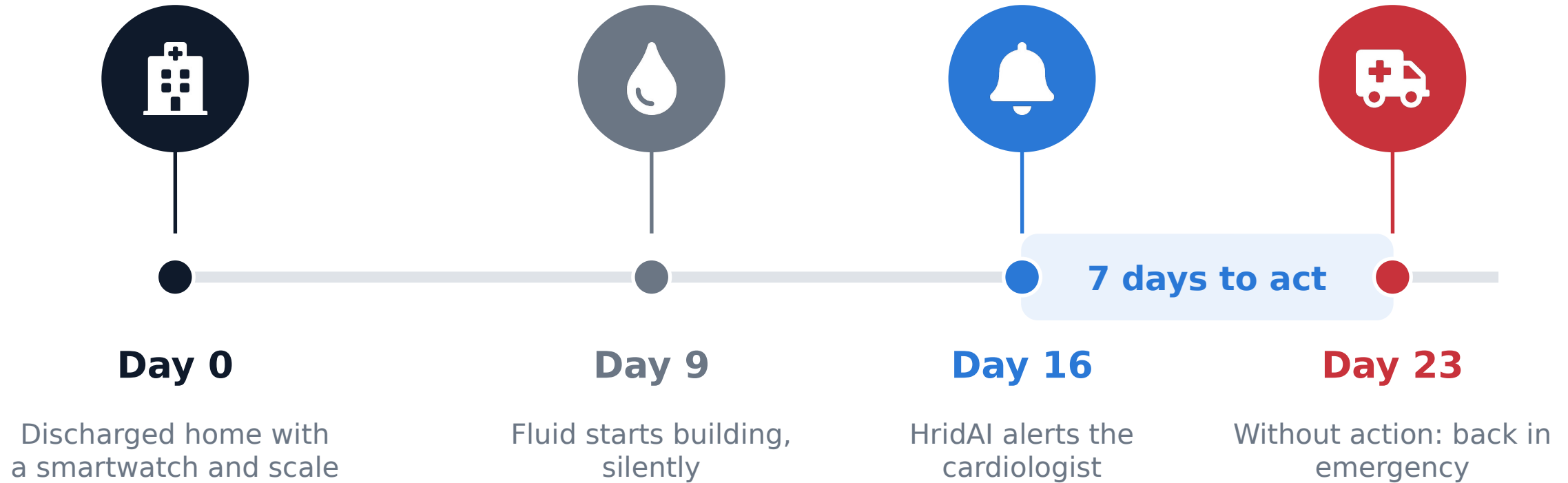
HridAI



A digital twin that sees heart failure coming.

Team Octas · Aditya Jain · PGDM (BM) 2026-28, XLRI Jamshedpur

Ramesh, 72, goes home after a heart-failure admission



Weight, night heart rate, HRV, breathing and sleep all shifted days earlier. Nobody was reading them together.

The month after discharge is when patients come back



~1 in 3

died within a year of a heart-failure admission

Trivandrum Heart Failure Registry, Kerala



~10 yrs

younger at presentation than patients in Western registries

Trivandrum Heart Failure Registry



6.5 days

median warning a wearable gave before rehospitalisation

LINK-HF study, Circ Heart Fail 2020

Today's home tool is a single weight rule

CURRENT GUIDANCE

Call your doctor if weight rises more than

2 kg in 3 days

58%

of readmissions caught in time in our test



One noisy signal

Daily weight swings by a kilo with food, clothes and scales.



Often skipped

Elderly patients miss weigh-ins, so the rule never sees the rise.



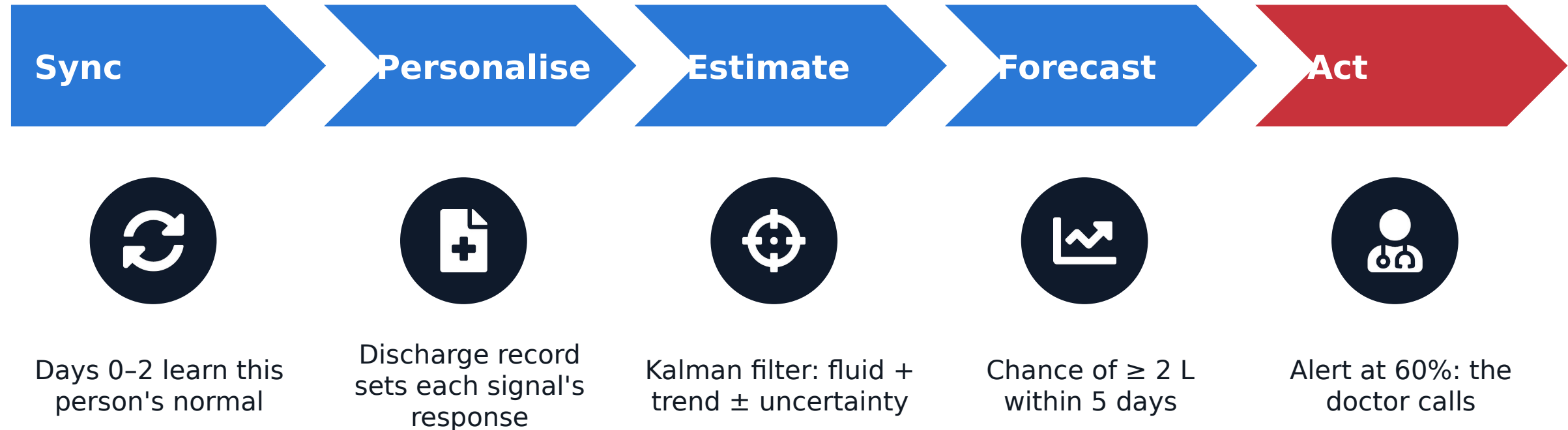
Ignores the rest

Heart rate, HRV, breathing and sleep change too, and go unread.

Every sensor sees a little of the hidden fluid



How the twin thinks, every night



Why a state-space twin? Works with 30 days of data per patient · explains every alert · handles missing days · gives calibrated probabilities

Architecture: two data streams in, one virtual patient

HridAI · system architecture

Heart-failure digital twin for the 30 days after hospital discharge · Team Octas, XLRI Jamshedpur · Happiest Health Digital Twin Challenge 2026

 Built in this proof-of-concept  Production roadmap

1 · DATA STREAMS

Static: EHR / discharge record

Age, sex, ejection fraction, eGFR, NT-proBNP, sodium, diabetes, hypertension, furosemide dose, prior HF admissions, polygenic risk percentile (genetic marker). *Synthetic, generated in `twin.py`.*

Dynamic: smartwatch (nightly)

Night heart rate, HRV (RMSSD), breathing rate, SpO2, sleep efficiency, steps. 10% missing days.

Dynamic: smart scale (daily)

Morning weight, with scale/clothing offsets and 25% skipped days.

Roadmap sources

Synthea FHIR bundles, MIMIC-IV, ABHA-linked records via ABDM; Apple Health / Google Fit / Health Connect; 3-tap WhatsApp symptom check.

2 · INGEST

Clean & align

One row per patient-day; missing values kept as gaps, never imputed.

EHR → twin priors

EF sets heart-rate response to fluid; EHR risk score sets how quickly the twin believes a rising trend.

LLM extraction

Free-text discharge summary → FHIR (Patient, Condition, Observation, MedicationStatement).

Consent & privacy

ABDM consent manager; DPDP Act-compliant storage; de-identified analytics.

3 · PATIENT DIGITAL TWIN (PER PATIENT)

- 1 **Sync** · days 0-2 at home learn the person's own baseline for every sensor.
- 2 **State estimation (Kalman filter)** · hidden state = [excess fluid L, trend L/day] with uncertainty. Every sensor is a noisy view of that one state.
$$x_{t+1} = F x_t + w \quad z_t = \text{baseline} + H(\text{EHR}) x_t + v$$
- 3 **5-day forecast** · $P(\text{fluid} \geq 2 \text{ L within 5 days})$, closed form from the projected mean and variance. Alert at $\geq 60\%$.
- 4 **Twin fidelity** · each day the twin predicts the next readings and scores how often reality lands inside its own 95% band.
- 5 **Active sensing** · uncertain + rising + scale skipped → ask the family for a weigh-in.
- 6 **What-if simulation** · project fluid for 10 days under a clinician-chosen diuretic change.

4 · PEOPLE WHO ACT

Clinician dashboard (Streamlit)

Belief band, risk, fidelity, alerts vs the 2 kg rule, EHR panel, day-by-day replay, what-if panel.

Caregiver message

Plain language, Hindi + English (templates in PoC).

Delivery

WhatsApp / SMS / IVR in Indian languages; nurse call-list for heart-failure clinics; EHR write-back as FHIR Observation.

Safety

Decision support only. The twin never changes treatment; the clinician decides.

Evaluation pipeline (`make_figures.py`) · dev cohort (seed 7) picks the alert threshold; untouched test cohort (seed 11, 500 patients) reports sensitivity 2-14 days ahead, lead time, false alerts per patient-month vs the 2 kg / 3 day guideline rule; ablations by data stream.

Synthetic cohort realism · person-specific sensor responses, fast and slow decompensators, salty-meal fluid bumps, scale offsets, missing days, and other-cause readmissions the twin should not claim.

Stack · Python, NumPy, pandas, Plotly, Streamlit, Matplotlib, pytest (10 tests incl. no look-ahead leakage). MIT licence. Runs on a laptop in under a minute.

Three things that make it a real twin



Twin fidelity

Every night it predicts the readings, then scores how many landed inside its own 95% band.



Active sensing

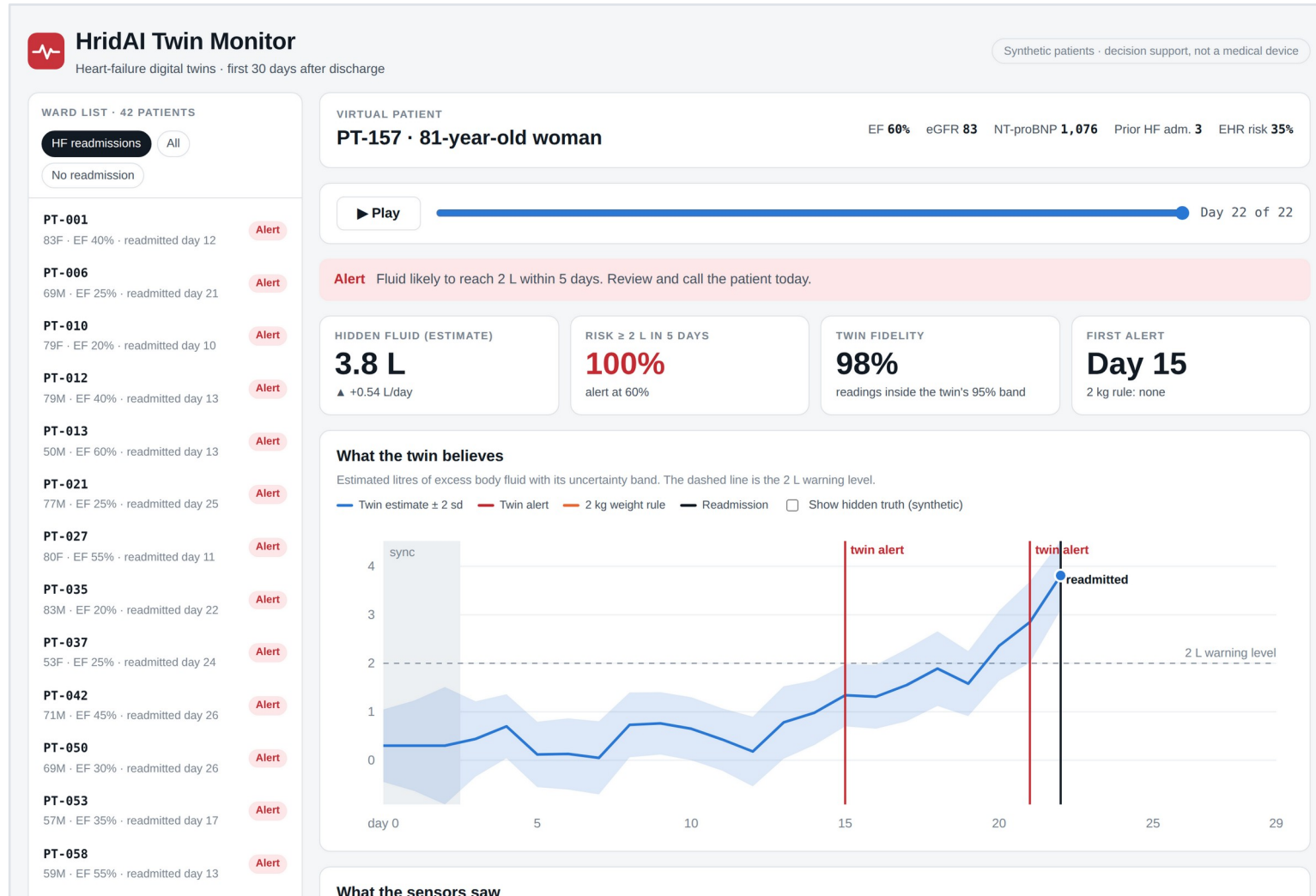
Unsure, rising, and the scale was skipped? It asks the family for a weigh-in.



What-if

The clinician tests a diuretic change on the twin before making it. The twin never acts alone.

The doctor's view: a live, working twin

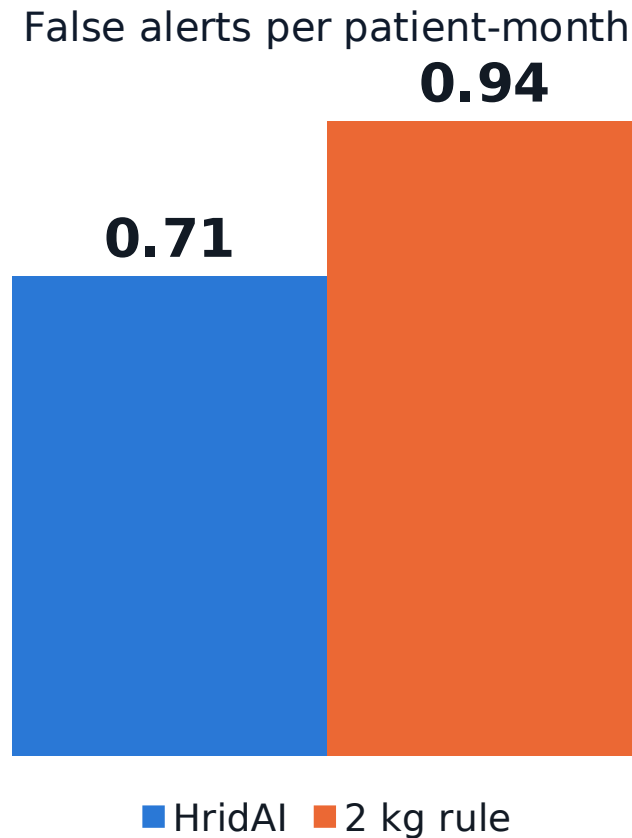
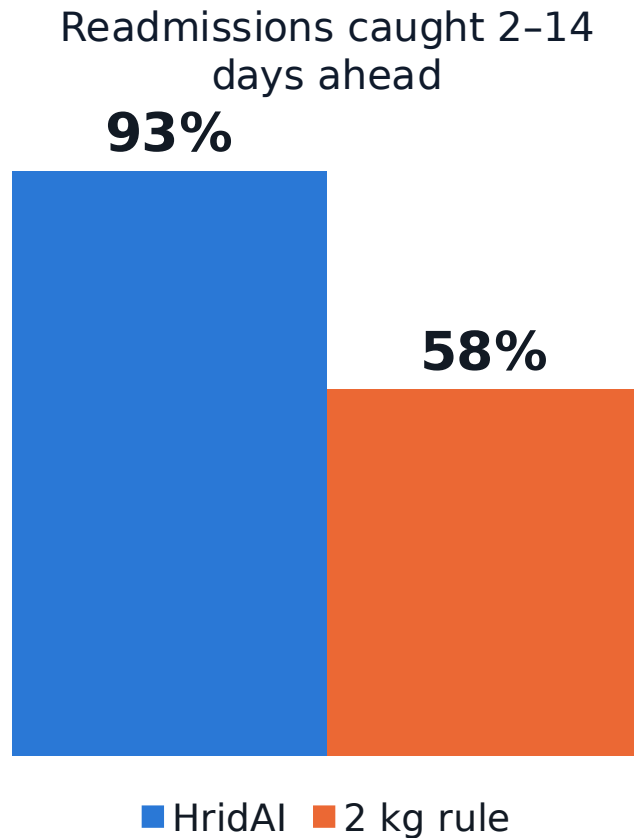


- Replay any patient day by day
- Fluid estimate with uncertainty
- Alerts vs the 2 kg rule
- Seven sensor trends
- What-if treatment panel
- Hindi and English family messages

 [Open live dashboard](#)

Runs in any browser. No install.

HridAI catches more readmissions, with fewer false alarms



4 days

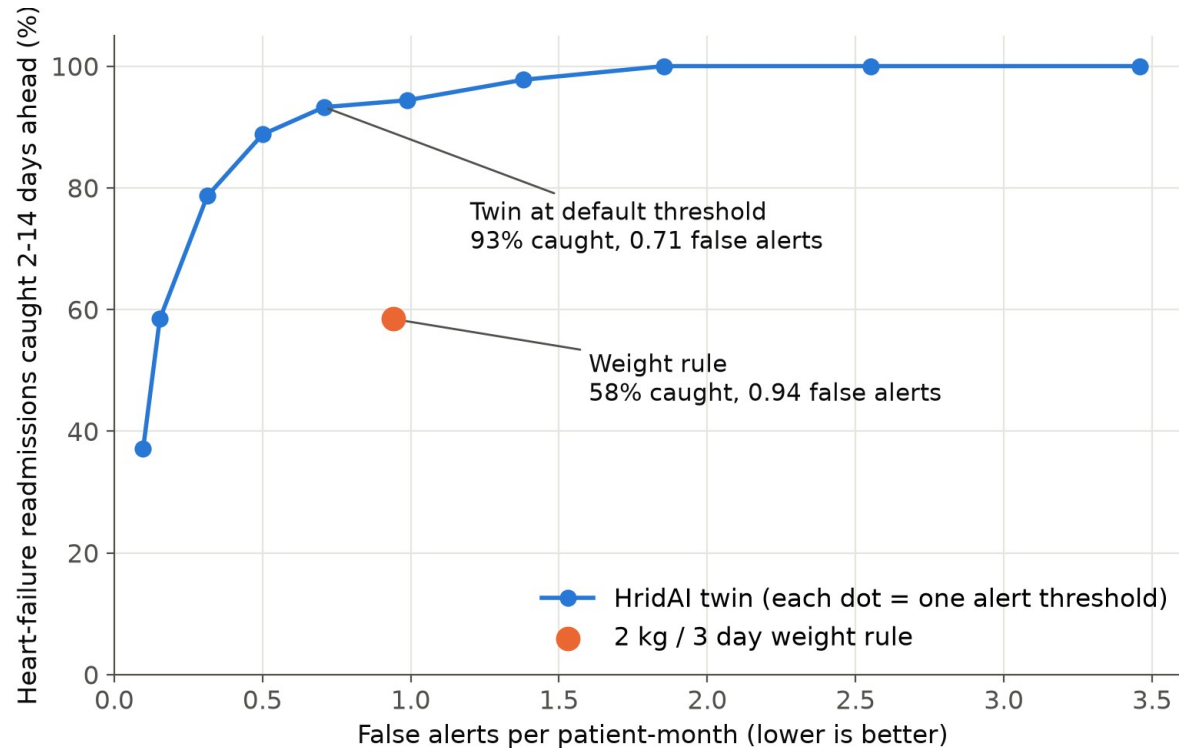
median warning before readmission

500

synthetic test patients, 89 heart-failure readmissions

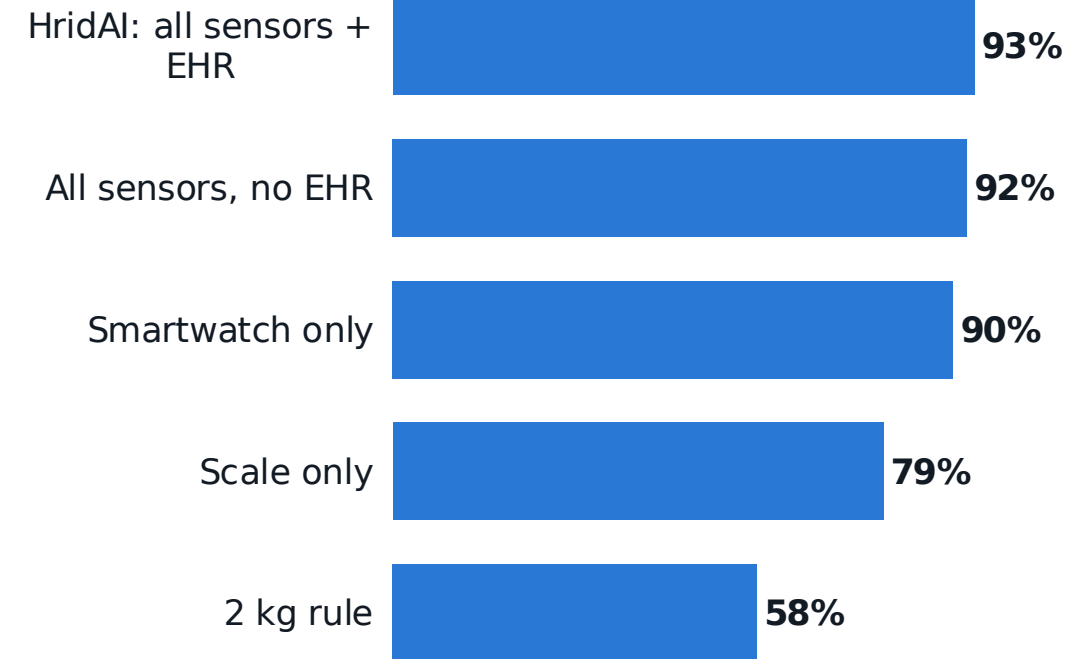
Alert threshold chosen on a separate development cohort. Synthetic data, so real-world performance needs clinical validation.

Every setting beats the rule; fusing streams drives it

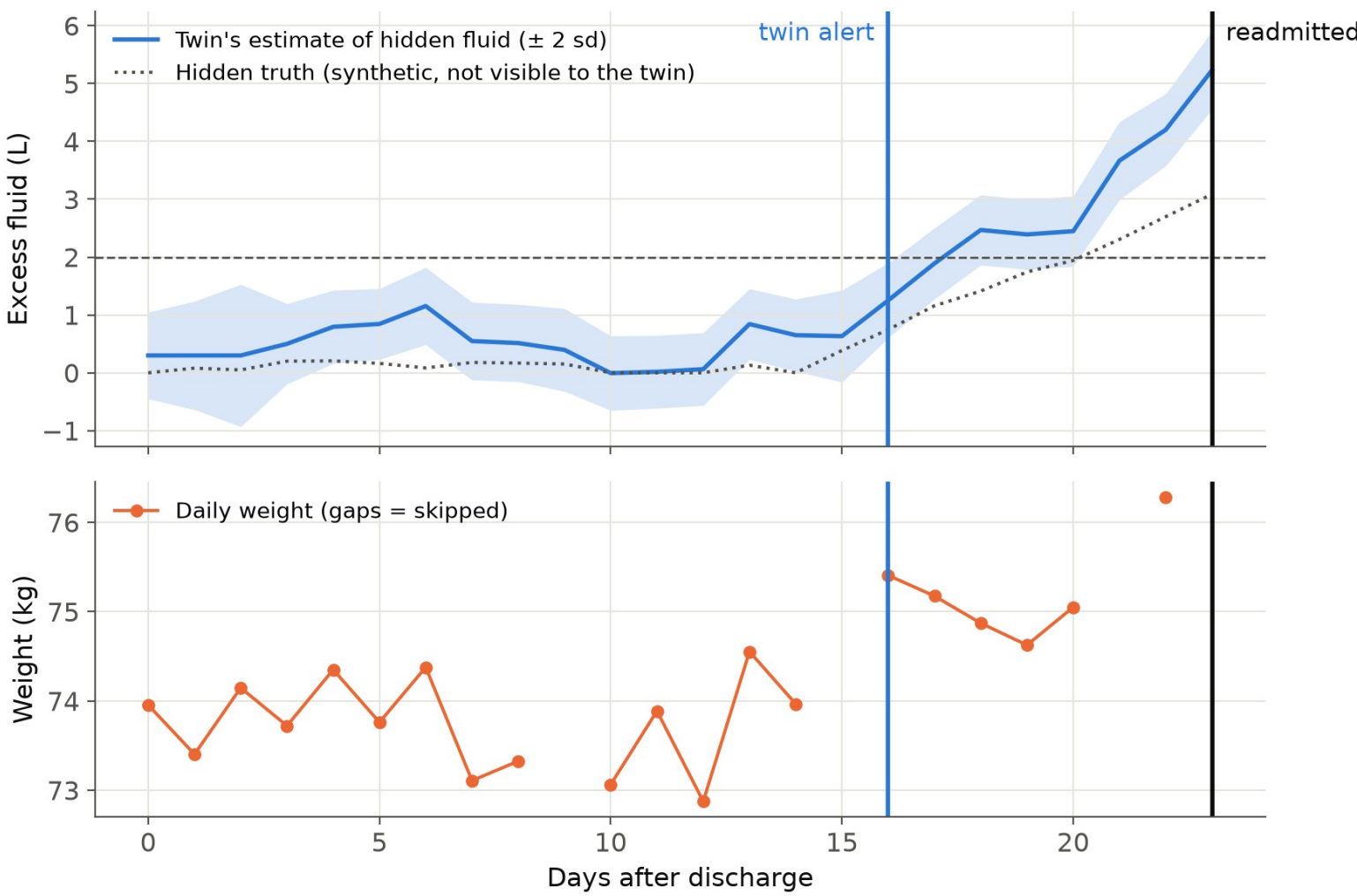


Each blue dot is a different alert threshold, so a clinic can choose more warning or fewer alarms.

What each data stream adds (readmissions caught)



One test patient: a week of warning the rule never gave



Day 16

HridAI alert

Day 23

readmitted

Never

the 2 kg rule fired

Synthetic data, built to be hard on the twin



Every body differs

Sensor responses vary per patient



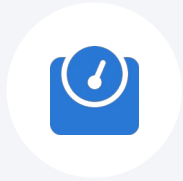
Fast crashes

Some decompensate in 4-5 days



Salty-meal bumps

Fluid spikes that fix themselves



Scale offsets

Clothes, different scales



Missing data

25% weigh-ins, 10% watch nights



Other readmissions

Causes the twin must not claim

EHR: age, sex, EF, eGFR, NT-proBNP, sodium, comorbidities, diuretic dose, prior admissions, polygenic risk ·
Wearables: 7 nightly / daily signals

Built for Indian homes and clinics



< ₹5,000 kit

Consumer smartwatch + Bluetooth scale, reusable



ABHA / ABDM

Records and consent on India's health rails



Indian languages

Family messages over WhatsApp or SMS

PILOT PLAN

1

cardiology ward

100

consecutive heart-failure discharges

30

days of monitoring each

1

primary outcome: 30-day readmissions vs matched controls

Who pays: hospitals (fewer readmissions), insurers (avoided claims), clinic networks (remote-monitoring service).

What the prototype does not yet prove, and what comes next

LIMITS

- Synthetic data: results are optimistic
- Sensor responses assumed, not fitted to Indian patients
- What-if is illustrative, not pharmacology
- Not a medical device: needs ethics approval and CDSCO review

NEXT STEPS

- Import Synthea FHIR records and open wearable data
- Learn each patient's sensor responses online
- Read free-text discharge summaries into the EHR
- Prospective pilot with a partner hospital



See heart failure coming. Act days earlier.



[Try the live twin](#)

Aditya Jain · Team Octas · XLRI Jamshedpur
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